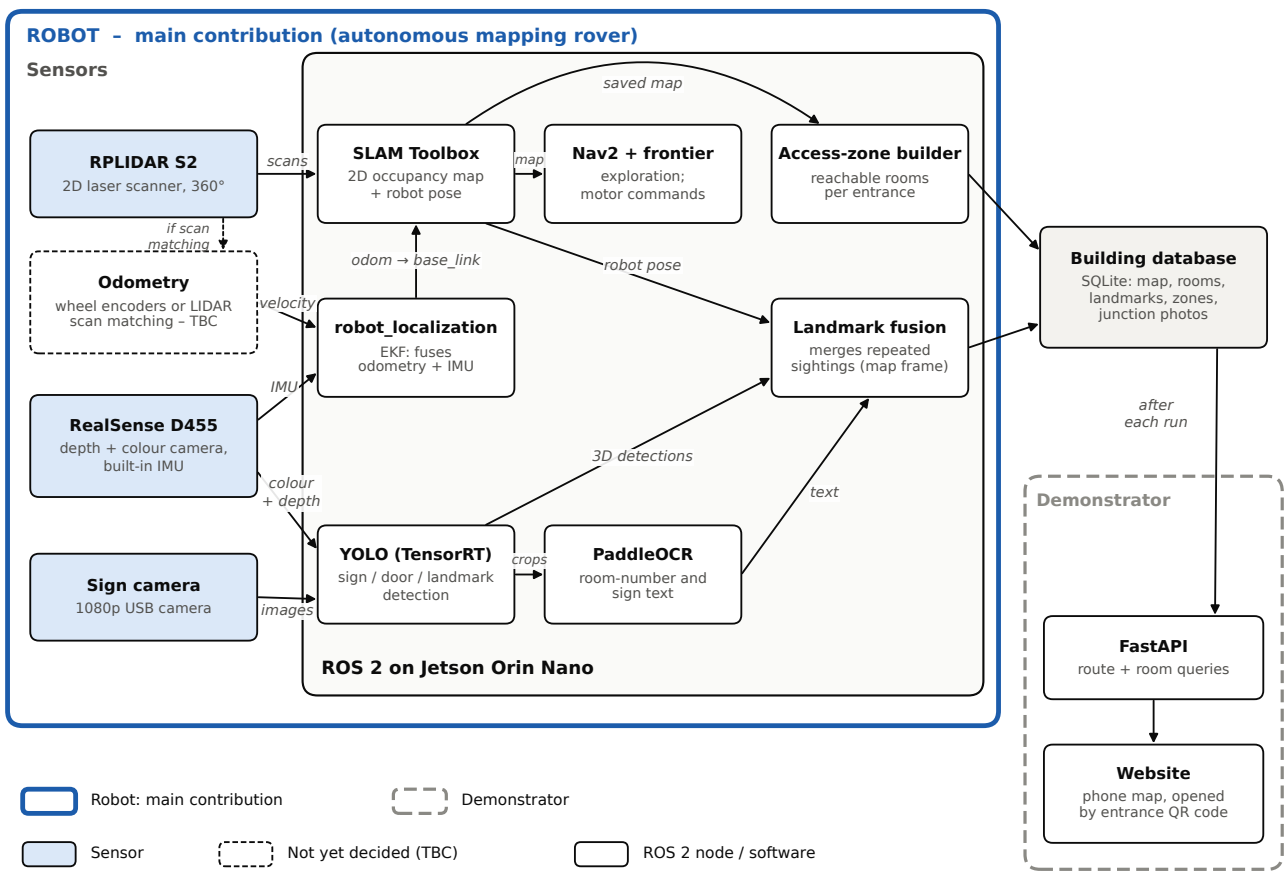
ROBOT – main contribution (autonomous mapping rover)
Sensors
RPLIDAR S2
2D laser scanner, 360°
scans
if scan matching
Odometry
wheel encoders or LIDAR scan matching – TBC
velocity
RealSense D455
depth + colour camera, built-in IMU
IMU
colour + depth
Sign camera
1080p USB camera
images
SLAM Toolbox
2D occupancy map + robot pose
saved map
map
Nav2 + frontier
exploration; motor commands
Access-zone builder
reachable rooms per entrance
odom → base_link
robot_localization
EKF: fuses odometry + IMU
robot pose
Landmark fusion
merges repeated sightings (map frame)
3D detections
YOLO (TensorRT)
sign / door / landmark detection
crops
PaddleOCR
room-number and sign text
text
ROS 2 on Jetson Orin Nano
Building database
SQLite: map, rooms, landmarks, zones, junction photos
after each run
Demonstrator
FastAPI
route + room queries
Website
phone map, opened by entrance QR code
Robot: main contribution
Demonstrator
Sensor
Not yet decided (TBC)
ROS 2 node / software